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Lab 07.2: Cromodora Seminar

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Abstract

This report summarizes the Cromodora seminar on the manufacturing, features, and quality planning of aluminum alloy wheels.

1. **Manufacturing Processes**
   - Cast Wheels: casting is common for its cost-effectiveness and styling flexibility. Processes include Gravity Die Casting (GDC) and Low-Pressure Die Casting (LPDC), the latter requiring precise cooling to prevent porosity
   - Forged Wheels: these offer superior strength and lower weight but at a higher cost. Techniques include Net Forging for high-volume production and Blank Forging/Machining for low-volume supercar applications
2. **Design and Quality Framework**
   - Key Features: wheels must meet strict requirements for geometry (compliance with hub design), interfaces (brake and tire clearance), aesthetics (finishes), and dynamics (fatigue and impact testing)
   - Safety Engineering: rims are designed with controlled failure modes to protect the suspension and undergo metallurgical testing to verify grain size and integrity
   - APQP Management: product development follows a five-phase Advanced Product Quality Planning framework, moving from feasibility and structural validation (FEA/DFMEA) to pre-series approval and full production

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3. Introduction

This report summarizes the key topics presented during a seminar delivered by Cromodora, focusing on the specialized manufacturing processes, the features and the testing of a wheel. Furthermore, some details on the production management and Advanced Product Quality Planning have been provided.

4. Manufacturing processes

Aluminium wheels are broadly divided into two main categories based on the manufacturing process: cast and forged. The selected technology depends on material characteristics, complexity of the product and overall cost.

4.1 Cast wheels

Casting processes are the most common methods, providing a large number of styling possibilities at a reasonable cost.

4.1.1 Gravity Die Casting (GDC):

- this is the simpler of the two casting methods, where the molten aluminium is poured into the die and fills the cavity primarily by gravity
- solidification is driven by the cooling system designed into the cast part

4.1.2 Low-Pressure Die Casting (LPDC):

- the molten metal fills the die from the bottom of the die through a rise tube which dips into a lower holding furnace which is pressurised with air to push and hold the metal in the die
- the hub typically solidifies last (in general, cooling proceeds in opposite direction compared to material injection). Careful control of the cooling front is crucial to prevent porosity formation (the speed of the solidification needs to be low to avoid the formation of porosities and cavities

4.2 Forged wheels

Forged wheels offer superior material characteristics and lower weight, often resulting in higher costs and constraints on styling complexity.

4.2.1 Net Forging (Low Cost/High Volume):

- used for standard fit, basic, and lightweight car specifications
- the process involves starting with a cast bar, followed by preliminary forging, design forging, perforation, flow-forming, and final finishing. Net forging limits the variety of stylings

4.2.2 Blank Forging and Machining (High Cost/Low Volume):

- used for supercars and hypercars due to the high costs involved

- the process involves converting a 6061 T6 billet into a forged blank, which is then finished-forged, spin-forged, and finally undergoes extensive milling to achieve elaborate stylings and overcome draft angle limitations

5. Wheel main features

5.1 Geometrical Features

- in general, the geometrical features are defined by the car manufacturer and the wheel manufacturer needs to comply with the requests
- number of holes is considered a challenge: on one side a high number of holes is better for casting operations; on the other hand, the car manufacturer who designs the wheel hub decides how many holes are there

5.2 Interface Features

- the wheel must ensure clearances for tyre types and sizes, brake caliper profiles and brake ventilation area
- the wheel must also be compatible with the hub diameter, with the position of the inflating valve, and bolt holes geometry

5.3 Cosmetics Features

- aesthetic requirements like colours and finishes are extremely important for the car manufacturer
- fully painted or milled/turned wheels are produced

5.4 Dynamic Features:

- lateral and radial loads are considered
- impact and fatigue tests are performed
- natural frequencies of the rim are important for NVH studies

5.5 Functional Features

- these focus on defining a limit for max unbalance and checking lateral runout
- definition of which tests are going to be performed on the rim and the requirements: mechanical, metallurgical, paint, functional tolerances

6. APQP

The systematic launch of a new wheel model is governed by the Advanced Product Quality Planning (APQP) framework, which organizes all quality and design activities into five sequential phases.

APQP Phase	Focus/responsibility	Keywords
Phase 1	project feasibility/sales responsibility	feasibility/cost analysis, timing plan
Phase 2	product design and dimensioning/sales and programme manager	CAD modeling, FEA for structural validation, DFMEA (Design Failure Mode and Effects Analysis), casting simulation
Phase 3	process fine-tuning/customer engineer	product tooling, sampling, mechanical tests
Phase 4	pre-series and production process approval/customer quality	quality studies, analysis of production performance
Phase 5	full production capacity and hand-over/programme manager	full capacity installation, production ramp up

Further considerations made about APQP phases:

- During the design phase it is a back and forth between customer and designer because the design might have to be changed to have good structural performances
- DFMEA is a list of the concerns and things that must be kept under control
- Wheelrim must be designed to break instead of damaging the suspension which is much worse
- Failure mode must be designed carefully
- Paint resistance to stone impacts must be tested
- Metallurgical test needs to be performed (micrography analysis) to verify if the solidification is well driven and to verify the presence of porosities; grain size analysis
- Production has to start within 1 year from the design presentation

7. Conclusion

The seminar provided a comprehensive overview of the aluminum alloy wheel rim industry, highlighting the main characteristics and differences between cast and forged (Net, Blank) wheels. The discussion underlined the importance of designing the wheel to meet strict geometrical, interface, cosmetic, dynamic, and functional features. Crucially, the entire product lifecycle is governed by the five-phase APQP framework.